ODE - 5

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Bernoulli's Equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Let $v = y^{1-n}$ be the substitution and continue

Let us try with an example.

(1) $\frac{dy}{dx} + \frac{y}{x} = y^2$

Divide throughout by y^2

$$y^{-2}\frac{dy}{dx} + y^{-1}x^{-1} = 1 \quad \text{—(1)}$$

$v = y^{1-2} = y^{-1}$

$\frac{dv}{dx} = -\frac{1}{y^2}\cdot\frac{dy}{dx}$

$\Rightarrow$ Putting in (1),

$$-\frac{dv}{dx} + \frac{v}{x} = 1 \quad \text{—(2)} \Rightarrow \frac{dv}{dx} - \frac{v}{x} = -1$$

$I.F = e^{-\int \frac{dx}{x}} = e^{-\ln x} = \frac{1}{x}$

I.F x (2) $\Rightarrow$

$$-\frac{1}{x}\cdot\frac{dv}{dx} + \frac{v}{x}\cdot\frac{1}{x} = \frac{1}{x}$$

$$\Rightarrow \frac{d}{dx}\left(-\frac{v}{x}\right) = \frac{1}{x}$$

Integrating $\Rightarrow -v/x = \ln x + c$

$\Rightarrow -v = x\ln x + cx$

$\Rightarrow -\frac{1}{y} = x\ln x + cx$

$\Rightarrow \frac{1}{y} + x\ln x + cx = 0$

(2) $-\frac{dy}{dx} + y = y^2 \quad \text{—(1)}$

Divide (1) with y^2,

$$-y^{-2}\cdot\frac{dy}{dx} + y^{-1} = 1 \quad \text{—(2)}$$

$y^{-1} = v \Rightarrow -\frac{1}{y^2}\cdot\frac{dy}{dx} = \frac{dv}{dx}$

(2) $\Rightarrow \frac{dv}{dx} + v = 1$

$\Rightarrow \frac{d}{dx}(vx) = 1$

$\Rightarrow vx = x + c$

$\Rightarrow v = 1 + \frac{c}{x}$

$\Rightarrow \boxed{\frac{1}{y} = 1 + \frac{c}{x}}$

(3) $\frac{dy}{dx} + \frac{2y}{x} = x^3y^2 \quad \text{—(1)}$

Divide (1) with y^2

$$y^{-2}\cdot\frac{dy}{dx} + 2y^{-1}x^{-1} = x^3$$

Put $v = y^{-1} \Rightarrow \frac{dv}{dx} = -\frac{1}{y^2}\cdot\frac{dy}{dx}$

$$\Rightarrow -\frac{dv}{dx} + \frac{2}{x} = x^3$$

$$\Rightarrow \frac{dv}{dx} - \frac{2}{x} = -x^3 \quad \text{—(2)}$$

We can find the I.F now

$I.F = e^{\int -2/x\,dx} = e^{-2\ln x}$

$= \frac{1}{x^2}$

Multiplying I.F to both sides of (2),

$$\frac{d}{dx}\left(v/x^2\right) = -x$$

Multiplying I.F to both sides of (a),

$\frac{d}{dx}(v/x^2) = -x$

Integrating,

$v/x^2 = -x^2/2 + C$

$\boxed{\frac{1}{x^2 y} + \frac{x^2}{2} = C}$

(4) $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

Divide by y^3 throughout,

$xy^{-2} - \frac{dy}{dx} \cdot y^{-3} = e^{-x^2}$ — (1)

$v = y^{-2} \Rightarrow \frac{dv}{dx} = -2/y^3 \cdot \frac{dy}{dx}$

(1) $\Rightarrow$ $xv + \frac{1}{2} \cdot \frac{dv}{dx} = e^{-x^2}$

$\Rightarrow \frac{dv}{dx} + 2vx = 2e^{-x^2}$ — (2)

I.F $= e^{\int 2x dx} = e^{x^2}$

(2) $\times$ I.F $\Rightarrow$ $\frac{d}{dx}(v \cdot e^{x^2}) = 2$

Integrating $\Rightarrow$ $v \cdot e^{x^2} = 2x + C$

$\Rightarrow \boxed{e^{x^2} = y^2(2x + C)}$

(5) $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$

Divide throughout by $\sec y$

$\cos y \cdot \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x)e^x$ — (1)

Let $v = \sin y$

$\frac{dv}{dx} = \cos y \cdot \frac{dy}{dx}$

(1) $\Rightarrow$ $\frac{dv}{dx} - \frac{v}{1+x} = (1+x)e^x$ — (2)

I.F $= e^{\int -\frac{1}{1+x} dx} = e^{-\ln(1+x)} = \frac{1}{1+x}$

I.F $\times$ (2) $\Rightarrow$ $\frac{1}{1+x} \cdot \frac{dv}{dx} - \frac{v}{(1+x)^2} = e^x$

$\Rightarrow \frac{d}{dx}\left(\frac{v}{1+x}\right) = e^x$

Integrating, $v = (1+x)[e^x + C]$

$\Rightarrow \sin y = (1+x)(e^x + C)$

(6) $\frac{dx}{dy} - (2y)x = y^3 x^2$

Divide throughout by x^2,

$x^{-2} \cdot \frac{dx}{dy} - x^{-1}(2y) = y^3$ — (1)

Let $z = x^{-1} \Rightarrow \frac{dz}{dy} = -\frac{1}{x^2} \cdot \frac{dx}{dy}$

(1) $\Rightarrow$ $\frac{dz}{dy} + 2yz = -y^3$ — (2)

I.F $= e^{+\int 2y dy} = e^{+y^2}$

(2) $\times$ I.F $\Rightarrow$ $e^{-y^2} \cdot \frac{dz}{dy} + 2yz \cdot e^{+y^2} = -y^3 \cdot e^{+y^2}$

I.F: $e^{\int \cdots} = e$

(2) × I.F ⇒ $e^{-y^2}\frac{dz}{dy} + 2yz\cdot e^{+y^2} = -y^3\cdot e^{+y^2}$

$\frac{d}{dy}\left(z\cdot e^{+y^2}\right) = -y^3\cdot e^{+y^2}$

Integrating, $z\cdot e^{y^2} = -\int y^3\cdot e^{+y^2}dy + c$

Let $y^2 = u$

$2ydy = du$

$\therefore \int y^3\cdot e^{y^2}dy = \frac{1}{2}\int u\cdot e^u\cdot du$

$= \frac{1}{2}\left[e^u\cdot u - \int e^u\,du\right]$

$= \frac{1}{2}\left[ue^u - e^u\right]$

$= \frac{1}{2}\left[e^{y^2}(y^2-1)\right] + c$

$\therefore \frac{1}{x}\cdot e^{y^2} = -\frac{1}{2}e^{y^2}(y^2-1) + c$

$\frac{1}{x} = \frac{1}{2}(1-y^2) + ce^{-y^2}$

(7) Try out:- $y(2xy + e^x)\,dx - e^x dy = 0$

Variation of parameters

We have to put it in the form $\frac{dy}{dx} + P(x)y = Q(x)$ first- then follow the previous method.

(8) $\frac{dy}{dx} + 3y = y^2\cos x$

(9) $\frac{dy}{dx} + \frac{y}{x}\log y = \frac{y(\log y)^2}{x}$